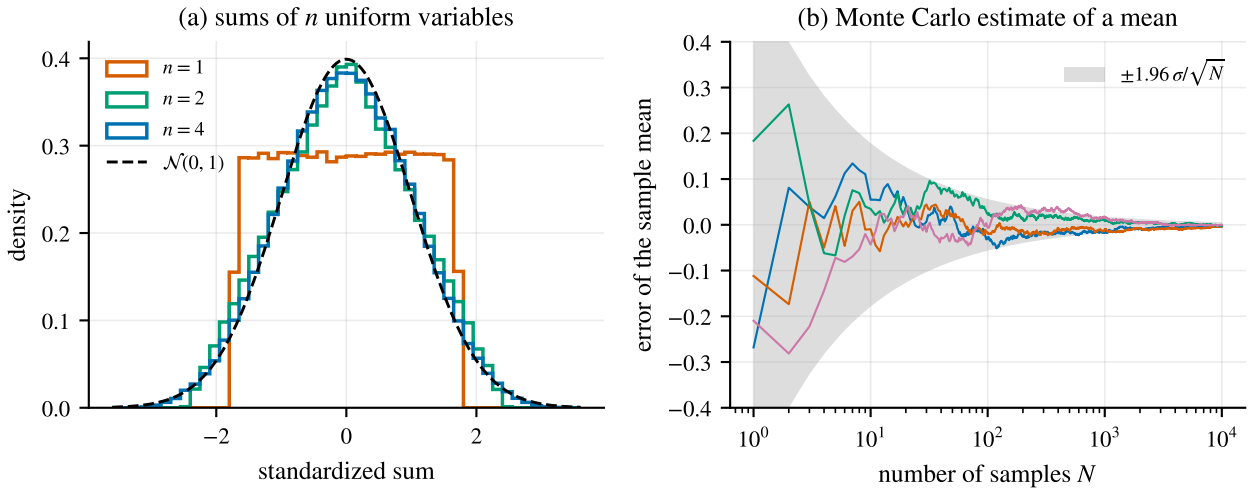

(a) sums of n uniform variables
density
standardized sum
$n = 1$
$n = 2$
$n = 4$
$\mathcal{N}(0, 1)$
0.4
0.3
0.2
0.1
0.0
−2
0
2
(b) Monte Carlo estimate of a mean
error of the sample mean
number of samples N
$\pm 1.96\,\sigma/\sqrt{N}$
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
10^0
10^1
10^2
10^3
10^4